Hall-Sensor Placement Geometry on a Conical Pole

Derivation of $\mathbf{p} = \mathbf{T} + s_{\mathrm{ax}}\mathbf{e}_2 + R(s_{\mathrm{ax}})\,\hat{\mathbf{r}} + a\,\mathbf{e}_3$

Scope. This note derives the closed-form construction used by `utils/pole_sensor_geometry.m`, which is the single source of truth for the six Hall-sensor centres $\mathbf{p}_i$ and their outward normals $\hat{\mathbf{n}}_i^+$. All vectors are expressed in the WP frame (origin at the common sphere centre of the six pole tips) in metres. The construction is per-pole; the index i is dropped.

1 Direction convention

Every direction below is written with the elevation–azimuth helper

$$\mathrm{dir}(\varepsilon,\theta) = \begin{bmatrix}\cos\varepsilon\cos\theta\\ \cos\varepsilon\sin\theta\\ \sin\varepsilon\end{bmatrix}, \qquad \mathrm{dir}(\varepsilon,\theta)\cdot\mathrm{dir}(\varepsilon',\theta) = \cos(\varepsilon-\varepsilon'). \tag{1}$$

The second identity is the only algebra needed to check orthogonality inside a meridian plane: two directions sharing the azimuth θ are orthogonal iff their elevations differ by 90°.

Per-pole inputs: azimuth θ (pole angles $0, 180, 120, 300, 60, 240°$ for $P_1 \ldots P_6$) and the layer flag (lower / upper).

2 Pole tip and pole axis

Tip. The six tips lie on a sphere of radius $\ell = R_{\mathrm{norm}}$. The magic-angle elevation follows from the locked hexapole constraints $R_{\mathrm{norm},xy} = \ell\sqrt{2/3}$, $R_{\mathrm{norm},z} = \ell/\sqrt{3}$:

$$\psi_0 = \arctan\frac{R_{\mathrm{norm},z}}{R_{\mathrm{norm},xy}} = \arctan\frac{1}{\sqrt{2}} = 35.2644°, \qquad \mathbf{e}_1 = \mathrm{dir}\big(\sigma_\ell\psi_0,\ \theta\big), \qquad \mathbf{T} = \ell\,\mathbf{e}_1, \tag{2}$$

with $\sigma_\ell = -1$ for a lower pole and $+1$ for an upper pole.

Axis. The cone axis $\mathbf{e}_2$ points from the tip *towards the base* (direction of increasing axial station s):

$$\mathbf{e}_2 = \mathrm{dir}(\varepsilon,\theta), \qquad \varepsilon = \begin{cases} 0 & \text{lower pole (horizontal axis)}\\ \varepsilon_{\mathrm{up}} = 36.5895° & \text{upper pole.}\end{cases} \tag{3}$$

Remark (do not confuse $\mathbf{e}_1$ with $\mathbf{e}_2$). The tip direction and the pole axis are *different* vectors. For a lower pole $\mathbf{e}_1$ has elevation $-35.2644°$ while $\mathbf{e}_2$ is horizontal; for an upper pole the elevations are $+35.2644°$ and $+36.5895°$. Only $\mathbf{e}_1$ carries the magic angle — it is fixed by the orthogonality/common-sphere constraints. The axis inclination $\varepsilon_{\mathrm{up}}$ is the physical tilt of the machined pole and is an independent number.

3 The local triad and the radial vector $\hat{\mathbf{r}}$

This is the part that is easy to forget. Fix a pole. Two more unit vectors are built so that $\{\mathbf{e}_2, \hat{\mathbf{r}}_0, \hat{\mathbf{t}}\}$ is orthonormal:

$$\hat{\mathbf{r}}_0 = \mathrm{dir}\big(\varepsilon + \sigma 90^\circ,\ \theta\big), \qquad \sigma = \begin{cases} -1 & \text{lower pole} \\ +1 & \text{upper pole} \end{cases} \tag{4}$$

$$\hat{\mathbf{t}} = \begin{bmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{bmatrix}. \tag{5}$$

Why these are orthonormal. $\mathbf{e}_2$ and $\hat{\mathbf{r}}_0$ share the azimuth θ and their elevations differ by exactly 90°, so $\mathbf{e}_2 \cdot \hat{\mathbf{r}}_0 = \cos(\mp 90^\circ) = 0$ by (1). The vector $\hat{\mathbf{t}}$ is horizontal and perpendicular to the azimuthal direction $[\cos\theta, \sin\theta, 0]^\mathsf{T}$, hence perpendicular to *any* $\mathrm{dir}(\cdot, \theta)$; in particular $\hat{\mathbf{t}} \perp \mathbf{e}_2$ and $\hat{\mathbf{t}} \perp \hat{\mathbf{r}}_0$.

Geometric meaning.

- $\mathbf{e}_2, \hat{\mathbf{r}}_0$ span the *meridian plane* — the vertical plane containing the pole axis (azimuth θ). $\hat{\mathbf{r}}_0$ is the radial direction inside that plane, pointing away from the axis.
- The sign σ selects *which side* of the axis the sensor is mounted on. A lower pole is the half-cut part: the mounting face is its **bottom** cone surface, so $\sigma = -1$ tilts $\hat{\mathbf{r}}_0$ downwards. An upper pole is a full cone and is mounted on the upper side, so $\sigma = +1$.
- $\hat{\mathbf{t}}$ is tangential — perpendicular to the meridian plane. It is only needed to rotate the mounting point around the pole axis.

Azimuthal placement around the axis. Rotating $\hat{\mathbf{r}}_0$ about $\mathbf{e}_2$ by an angle ψ stays in the plane spanned by $\hat{\mathbf{r}}_0$ and $\hat{\mathbf{t}}$ (both are perpendicular to $\mathbf{e}_2$), so Rodrigues' formula collapses to

$$\boxed{\hat{\mathbf{r}}(\psi) = \cos\psi\, \hat{\mathbf{r}}_0 + \sin\psi\, \hat{\mathbf{t}}} \tag{6}$$

with $\hat{\mathbf{r}}(\psi) \perp \mathbf{e}_2$ and $\|\hat{\mathbf{r}}(\psi)\| = 1$ for all ψ. The canonical sensor azimuth is $\psi = 0$, i.e. $\hat{\mathbf{r}} = \hat{\mathbf{r}}_0$, which places the sensor in the meridian plane — directly below (lower) or above (upper) the pole axis.

Caveat on the sense of rotation. A direct evaluation of the cross product gives $\mathbf{e}_2 \times \hat{\mathbf{r}}_0 = -\sigma \hat{\mathbf{t}}$. Therefore (6) is a right-handed rotation about $\mathbf{e}_2$ by $-\sigma\psi$: the sense of increasing ψ is opposite between the two layers. This is immaterial at the default $\psi = 0$ and for symmetric sweeps, but it must be kept in mind if a signed azimuthal offset is ever imposed.

4 The real cone: virtual apex, radius, and outward normal

4.1 Virtual apex

The pole is not a mathematically sharp cone: the tip is rounded with a fillet of radius $r_f = 40\ \mu$m, and the conical surface is *tangent* to that sphere. Let β be the true half-cone angle. For a sphere of radius r_f inscribed tangentially in a cone of half-angle β, the distance from the apex to the sphere centre is $r_f / \sin\beta$. Placing the sphere centre at axial station $s = r_f$ (so that the foremost point of the sphere — the physical tip $\mathbf{T}$ — sits at $s = 0$), the apex lies *outside* the material at

$$s = -e_v, \qquad \boxed{e_v = \frac{r_f}{\sin\beta} - r_f} \approx 0.16\ \text{mm}. \tag{7}$$

4.2 Cone radius

Measuring the axial station s from the physical tip, the distance from the virtual apex is $e_v + s$, so

$$R(s) = (e_v + s)\tan\beta, \qquad R(0) = e_v \tan\beta \approx 0.033 \text{ mm} \neq 0. \tag{8}$$

The non-zero intercept is the whole point: **at the axial position of the tip the cone surface already has a finite radius**. Anchoring the mounting point at the ideal tip with $R = 0$ is what used to push the sensor into the steel (Sec. 8).

4.3 Outward normal

Parametrise the cone surface by (s, ψ):

$$\mathbf{P}(s,\psi) = \mathbf{T} + s\,\mathbf{e}_2 + R(s)\,\hat{\mathbf{r}}(\psi). \tag{9}$$

The generator (slant) direction is

$$\hat{\mathbf{g}} = \frac{\partial \mathbf{P}/\partial s}{\|\partial \mathbf{P}/\partial s\|} = \frac{\mathbf{e}_2 + \tan\beta\,\hat{\mathbf{r}}}{\sqrt{1+\tan^2\beta}} = \cos\beta\,\mathbf{e}_2 + \sin\beta\,\hat{\mathbf{r}}. \tag{10}$$

The surface normal lies in the meridian-like plane $\mathrm{span}\{\mathbf{e}_2, \hat{\mathbf{r}}\}$, is perpendicular to $\hat{\mathbf{g}}$, and must point away from the axis ($\hat{\mathbf{n}} \cdot \hat{\mathbf{r}} > 0$). The unique unit vector satisfying all three is

$$\boxed{\mathbf{e}_3 \;=\; \hat{\mathbf{n}}^{+} \;=\; \cos\beta\,\hat{\mathbf{r}} - \sin\beta\,\mathbf{e}_2} \tag{11}$$

since $\hat{\mathbf{g}} \cdot \mathbf{e}_3 = \cos\beta\sin\beta - \sin\beta\cos\beta = 0$ and $\mathbf{e}_3 \cdot \hat{\mathbf{r}} = \cos\beta > 0$. At $\psi = 0$ this is exactly $\mathrm{dir}\big(\varepsilon + \sigma(\beta + 90^\circ), \theta\big)$, i.e. the axis direction rotated by $\sigma(90^\circ + \beta)$ inside the meridian plane.

5 Axial station from the slant offset

SOFF is defined as the **slant distance measured along the generator**, starting from the "red point" — the point of the generator at the same axial station as the tip, i.e. $(s, R) = (0, e_v \tan\beta)$. Advancing a slant distance SOFF along $\hat{\mathbf{g}}$ advances the axial station by its $\mathbf{e}_2$ component, hence

$$\boxed{s_{\mathrm{ax}} = \mathrm{SOFF} \cdot \cos\beta} \tag{12}$$

with the design value $\mathrm{SOFF} = 4.572$ mm ($= 0.18$ inch).

Convention. Quoting a mounting offset always means the *full* slant distance measured in the meridian plane, *including* its leading segment: saying "SOFF = 4 mm" already contains the portion t_{tan} that is pressed against the fillet rather than the cone, and the whole 4 mm is then projected onto the axis by $\cos\beta$. The CAD-measured fillet segments are $t_{\mathrm{tan}} = 0.0328$ mm (lower) and 0.0311 mm (upper); they are recorded only for drawing the pole outline and do *not* enter (12).

History. An earlier revision used $s_{\mathrm{ax}} = (\mathrm{SOFF} - t_{\mathrm{tan}})\cos\beta + t_{\mathrm{tan}}$, which measured SOFF from the physical tip and treated the leading segment as purely axial. The two differ by $t_{\mathrm{tan}}(1 - \cos\beta) \approx 0.6\ \mu$m, three orders of magnitude below the air gap, so no downstream result changed.

6 Assembly

Stacking (2), (12), (8), (6) and (11), and lifting the sensor off the surface by the air gap $a = 0.41$ mm along the outward normal:

$$\boxed{\mathbf{p} = \underbrace{\mathbf{T}}_{\text{tip}} + \underbrace{s_{\mathrm{ax}}\,\mathbf{e}_2}_{\text{slide along axis}} + \underbrace{R(s_{\mathrm{ax}})\,\hat{\mathbf{r}}(\psi)}_{\text{out to the cone surface}} + \underbrace{a\,\mathbf{e}_3}_{\text{lift off the surface}}} \tag{13}$$

Read left to right, the four terms are: start at the pole tip; slide along the axis to the correct axial station; step radially outwards until you reach the actual cone surface at that station; then lift perpendicular to that surface by the air gap. Because the last step uses the true surface normal (11), the perpendicular distance from the sensor centre to the cone surface is *exactly a* by construction.

Degenerate case: flat mounting face

For the lower pole the option `face_lower='flat'` mounts the sensor on the half-cut plane instead of the bottom cone. That plane *contains* the pole axis, so there is no radial stand-off and no slant projection:

$$s_{\mathrm{ax}} = \mathrm{SOFF}, \qquad R = 0, \qquad \hat{\mathbf{n}}^{+} = \hat{\mathbf{z}}, \qquad \mathbf{p} = \mathbf{T} + \mathrm{SOFF}\,\mathbf{e}_2 + a\,\hat{\mathbf{z}}. \tag{14}$$

The azimuth ψ has no meaning for a plane and is ignored.

7 Numerical values (long2016 half-cut, CAD-measured cone)

$\ell = 0.5$ mm, $r_f = 0.04$ mm, SOFF $= 4.572$ mm, $a = 0.41$ mm. The two layers have *different* measured cone slopes ($\tan\beta = 0.20330$ lower, 0.19463 upper; a 4.4% difference).

Layer	β [deg]	e_v [mm]	s_{ax} [mm]	$R(s_{\mathrm{ax}})$ [mm]	poles
Lower	11.4916	0.16078	4.48035	0.94354	P_1, P_3, P_6
Upper	11.0138	0.16937	4.48779	0.90642	P_2, P_4, P_5

Spot check for P_1 ($\theta = 0$, lower, $\psi = 0$): $\mathbf{e}_2 = [1,0,0]^{\mathsf{T}}$, $\hat{\mathbf{r}}_0 = \mathrm{dir}(-90^\circ, 0) = [0,0,-1]^{\mathsf{T}}$, so (11) gives

$$\hat{\mathbf{n}}^{+} = \cos\beta\,[0,0,-1]^{\mathsf{T}} - \sin\beta\,[1,0,0]^{\mathsf{T}} = [-0.1992,\ 0,\ -0.9800]^{\mathsf{T}},$$

which is what the code returns. The normal tilts β away from straight-down, as it must for a surface whose generator is inclined by β to the axis.

8 Why the naive construction fails

A tempting shortcut is to anchor at the ideal tip and walk the slant distance directly: $\mathbf{p} = \mathbf{T} + \mathrm{SOFF}\cdot\mathrm{dir}(\varepsilon + \sigma\beta, \theta) + a\,\mathbf{e}_3$. This ignores the finite radius $R(0) = e_v \tan\beta \approx 0.033$ mm at the tip station, so the mounting point is placed *inside* the steel. The resulting perpendicular distance to the *real* cone surface is

$$0.3635 \text{ mm (lower)}, \qquad 0.4012 \text{ mm (upper)},$$

instead of the specified 0.41 mm — an 11% error on the lower poles, and inconsistent between layers. The construction (13) removes both defects: the stand-off is exactly a on every pole.

Implementation: `matlab/Maxwell/utils/pole_sensor_geometry.m` (and the identical copy under `matlab/APDL/Calibratio`
Consumers: `build_V_matrix.m` and every script that draws sensors or pole outlines — none of them may re-derive this geometry locally.

Related: `.claude/rules/ansys-cad-alignment.md` (CAD is the source of truth); memory `long2016-pole-cone-geometry` (STEP measurements and how they were taken).